CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 3: Pair of Linear Equations in Two Variables

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Graphical method and consistency

1. The graph of a pair of linear equations in two variables is a pair of coincident lines. How many solutions does the pair have? (a) exactly one (b) two (c) none (d) infinitely many CBSE 2026, 1 mark

SOLUTION

Coincident lines lie on top of each other, so every point of the line is common to both.

Answer: (d)

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2. The lines $2x - 3y + 6 = 0$, $2x + 3y - 18 = 0$ and $x = 0$ form a triangle. Draw the graph, write the coordinates of its vertices, and find its area. CBSE 2026, 5 marks

SOLUTION

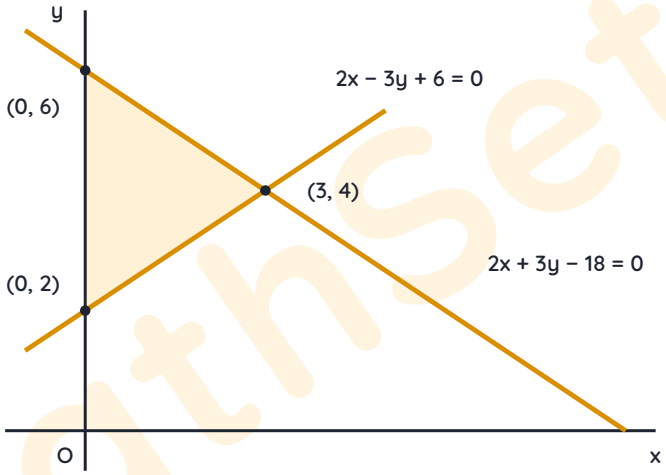
Points on $2x - 3y + 6 = 0$:

x	0	3	-3
y	2	4	0

Points on $2x + 3y - 18 = 0$:

x	0	3	9
y	6	4	0

The line $x = 0$ is the y-axis. Plot the points and join them.



The two slanting lines meet at $(3, 4)$. They meet the y-axis at $(0, 2)$ and $(0, 6)$.

Base along the y-axis $= 6 - 2 = 4$ units. Height $=$ distance of $(3, 4)$ from the y-axis $= 3$ units.

Area $= \frac{1}{2} \times 4 \times 3$

Answer: vertices $(0, 2)$, $(0, 6)$ and $(3, 4)$; area $= 6$ square units

3. A graph shows the lines of two linear equations. The lines are not parallel, but they do not meet in the part of the plane that is drawn. The pair of equations is: (a) consistent, with a unique solution (b) consistent, with infinitely many solutions (c) inconsistent (d) inconsistent, but made consistent by extending the lines CBSE 2024, 1 mark

SOLUTION

The two lines are not parallel, so when extended they meet at exactly one point.

A pair of lines that intersect at one point has a unique solution and is consistent.

Answer: (a)

4. For what value of k does the pair $kx = y + 2$ and $6x = 2y + 3$ have infinitely many solutions? (a) $k = 3$ (b) no such value exists (c) $k = -3$ (d) $k = 4$ CBSE 2023, 1 mark

SOLUTION

Write the pair as $kx - y - 2 = 0$ and $6x - 2y - 3 = 0$.

For infinitely many solutions we need $\frac{k}{6} = \frac{-1}{-2} = \frac{-2}{-3}$.

But $\frac{1}{2} \neq \frac{2}{3}$, so the condition can never be met.

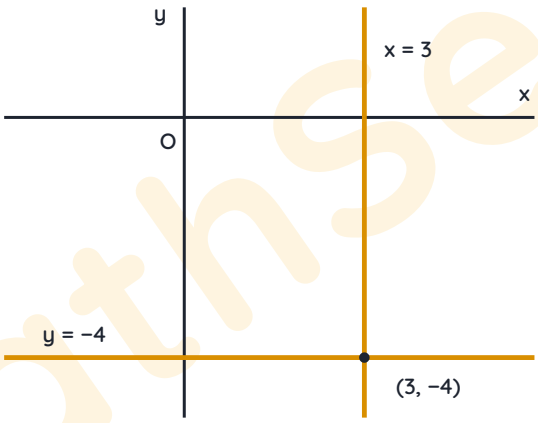
Answer: (b)

5. Solve the pair of equations $x = 3$ and $y = -4$ graphically. CBSE 2023, 2 marks

SOLUTION

The graph of $x = 3$ is a line parallel to the y-axis, 3 units to its right.

The graph of $y = -4$ is a line parallel to the x-axis, 4 units below it.



The two lines cross at one point.

Answer: $x = 3, y = -4$

6. Use the graphical method to decide whether the pair $x = 0$ and $y = -7$ is consistent. CBSE 2023, 2 marks

SOLUTION

The graph of $x = 0$ is the y-axis. The graph of $y = -7$ is a line parallel to the x-axis, 7 units below it.

These two lines are perpendicular, so they meet at exactly one point, $(0, -7)$.

Answer: the pair is consistent, with the unique solution $x = 0, y = -7$

7. Find c if the pair $cx + 3y + (3 - c) = 0$ and $12x + cy - c = 0$ has infinitely many solutions. CBSE 2019, 2 marks

SOLUTION

For infinitely many solutions, $\frac{c}{12} = \frac{3}{c} = \frac{3-c}{-c}$.

From the first two ratios, $c^2 = 36$, so $c = 6$ or $c = -6$.

From the last two ratios, $-3c = 3c - c^2$, so $c^2 = 6c$ and $c = 6$ (since $c \neq 0$).

Only $c = 6$ satisfies both conditions.

Answer: $c = 6$

2 Elimination method

8. The pair of equations $2x - 3y + a = 0$ and $2x + 3y - b = 0$ has the solution $x = 1$, $y = 2$. Find a and b, and state how they are related. CBSE 2025, 1 mark

SOLUTION

First equation: $2(1) - 3(2) + a = 0$, so $a = 4$.

Second equation: $2(1) + 3(2) - b = 0$, so $b = 8$.

Answer: $a = 4$, $b = 8$, **so** $b = 2a$

9. Solve the pair $7x - 2y = 5$ and $8x + 7y = 15$, and verify the solution. CBSE 2024, 2 marks

SOLUTION

Multiply the first equation by 7 and the second by 2: $49x - 14y = 35$ and $16x + 14y = 30$

Add: $65x = 65$, so $x = 1$. From $7x - 2y = 5$: $7 - 2y = 5$, so $y = 1$.

Verification: $7(1) - 2(1) = 5$ and $8(1) + 7(1) = 15$. Both are true.

Answer: $x = 1$, $y = 1$

10. $ABCD$ is a rectangle in which $AB = 30$ cm, $DC = (x + y)$ cm, $AD = 14$ cm and $BC = (x - y)$ cm. Find x and y. CBSE 2018, 2 marks

SOLUTION

Opposite sides of a rectangle are equal: $x + y = 30$ ... (1) and $x - y = 14$ ... (2)

Adding: $2x = 44$, so $x = 22$. From (1), $y = 8$.

Answer: $x = 22$, $y = 8$

3 Word problems

11. A person puts money into two schemes, A and B, which pay simple interest at 8% and 9% a year. The interest for one year comes to ₹1,860. If the two amounts had been swapped between the schemes, the interest would have been ₹20 more. How much was put into each scheme? CBSE 2025, 5 marks

SOLUTION

Let ₹x be invested in scheme A and ₹y in scheme B.

Interest as invested: $\frac{8x}{100} + \frac{9y}{100} = 1860$, so $8x + 9y = 186000$ ... (1)

Interest after swapping: $\frac{9x}{100} + \frac{8y}{100} = 1880$, so $9x + 8y = 188000$ ... (2)

Adding (1) and (2): $17(x + y) = 374000$, so $x + y = 22000$ ... (3)

Subtracting (1) from (2): $x - y = 2000$ ... (4)

Adding (3) and (4): $2x = 24000$, so $x = 12000$ and $y = 10000$.

Check: 8% of $12000 + 9\%$ of $10000 = 960 + 900 = 1860$.

Answer: ₹12,000 in scheme A and ₹10,000 in scheme B

12. Three years ago Rashmi's age was three times Nazma's age. Ten years from now Rashmi's age will be twice Nazma's. Find their present ages. CBSE 2024, 3 marks

SOLUTION

Let Rashmi be x years old and Nazma y years old now.

Three years ago: $x - 3 = 3(y - 3)$, so $x = 3y - 6$.

Ten years later: $x + 10 = 2(y + 10)$, so $x = 2y + 10$.

$3y - 6 = 2y + 10$ gives $y = 16$, and $x = 42$.

Answer: Rashmi is 42 years old and Nazma is 16 years old.

13. Half the difference of two numbers is 2. The larger number added to twice the smaller number gives 13. Find the two numbers. CBSE 2023, 3 marks

SOLUTION

Let the larger number be x and the smaller be y.

$\frac{x - y}{2} = 2$, so $x - y = 4$ ... (1)

$x + 2y = 13$ ... (2)

Subtracting (1) from (2): $3y = 9$, so $y = 3$ and $x = 7$.

Answer: 7 and 3

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14. A father is three times as old as the sum of the ages of his two children. In 5 years he will be twice as old as the sum of their ages. Find his present age. CBSE 2019, 3 marks

SOLUTION

Let the father be x years old and the sum of the children's ages be y years. Then $x = 3y$ ... (1)

After 5 years the father is $x + 5$ and the children together are $y + 10$: $x + 5 = 2(y + 10)$ ... (2)

Put (1) in (2): $3y + 5 = 2y + 20$, so $y = 15$ and $x = 45$.

Answer: 45 years

15. A fraction becomes $\frac{1}{3}$ when 2 is taken from its numerator, and becomes $\frac{1}{2}$ when 1 is taken from its denominator. Find the fraction. CBSE 2019, 3 marks

SOLUTION

Let the fraction be $\frac{x}{y}$.

$\frac{x-2}{y} = \frac{1}{3}$, so $y = 3x - 6$ ... (1)

$\frac{x}{y-1} = \frac{1}{2}$, so $y = 2x + 1$ ... (2)

$3x - 6 = 2x + 1$, so $x = 7$ and $y = 15$.

Answer: $\frac{7}{15}$

16. A boat covers 30 km upstream and 44 km downstream in 10 hours. In 13 hours it covers 40 km upstream and 55 km downstream. Find the speed of the stream and the speed of the boat in still water. CBSE 2019, 4 marks

SOLUTION

Let the upstream speed be u km/h and the downstream speed be v km/h. Put $a = \frac{1}{u}$ and $b = \frac{1}{v}$.

$30a + 44b = 10$ ... (1) and $40a + 55b = 13$ ... (2)

$4 \times (1)$: $120a + 176b = 40$. $3 \times (2)$: $120a + 165b = 39$. Subtracting, $11b = 1$, so $b = \frac{1}{11}$.

From (1): $30a = 10 - 4 = 6$, so $a = \frac{1}{5}$. Hence $u = 5$ and $v = 11$.

Boat speed $= \frac{v+u}{2} = 8$ km/h and stream speed $= \frac{v-u}{2} = 3$ km/h.

Answer: boat 8 km/h; stream 3 km/h

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